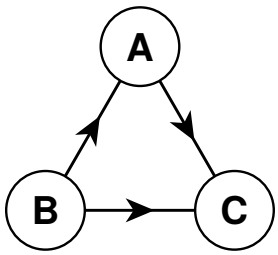
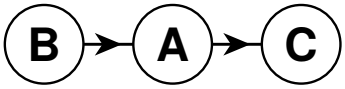
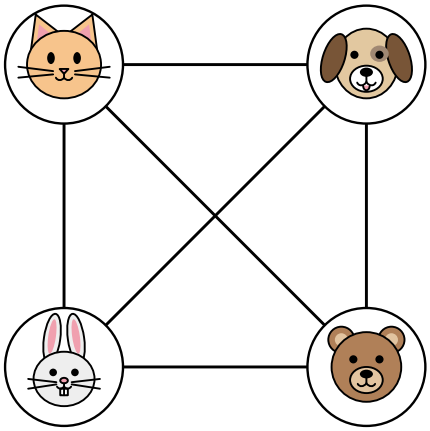
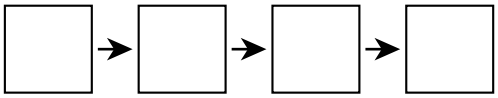
Every two players play one game, and one of them wins. An arrow points from the winner to the loser. A line-up is a row of all the players in which each player beat the next one.

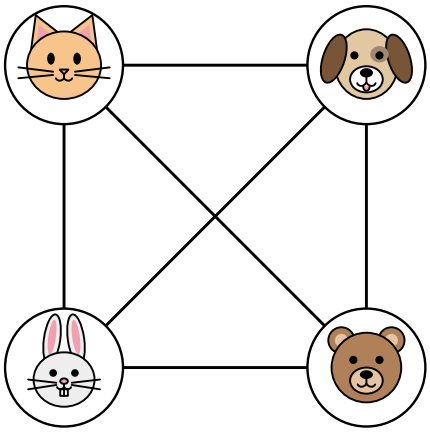


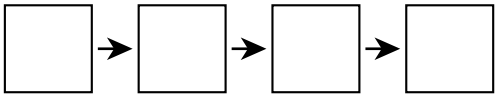


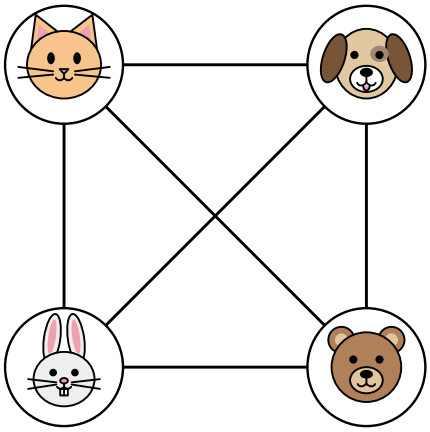
Problem 1: Play three tournaments with the cat, dog, bear and rabbit cards, deciding each game by rock-paper-scissors. Each time, draw the arrows on a picture below and find a line-up.

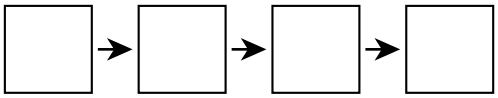




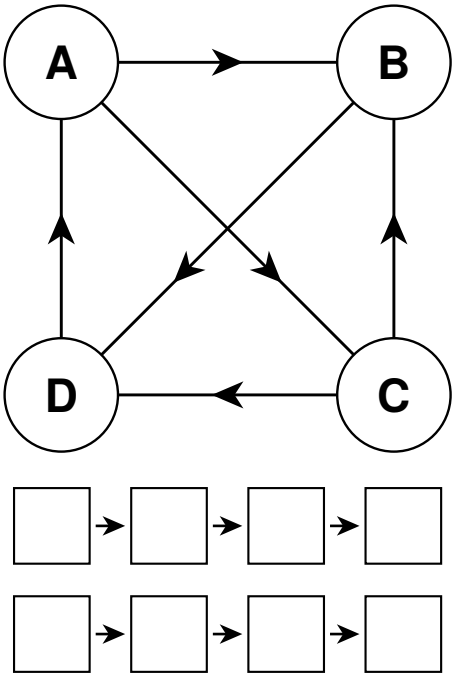


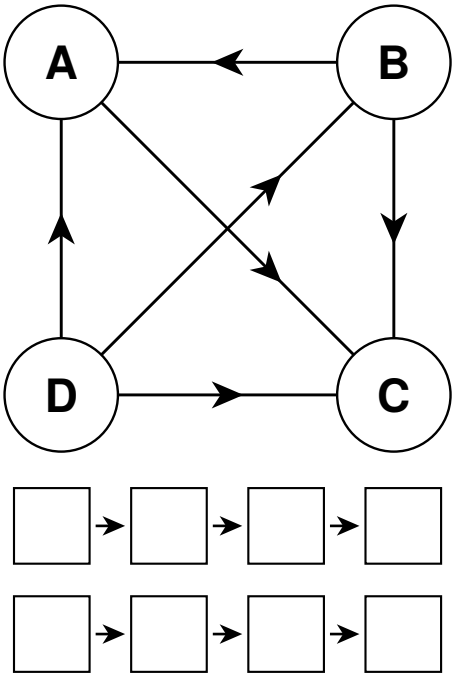


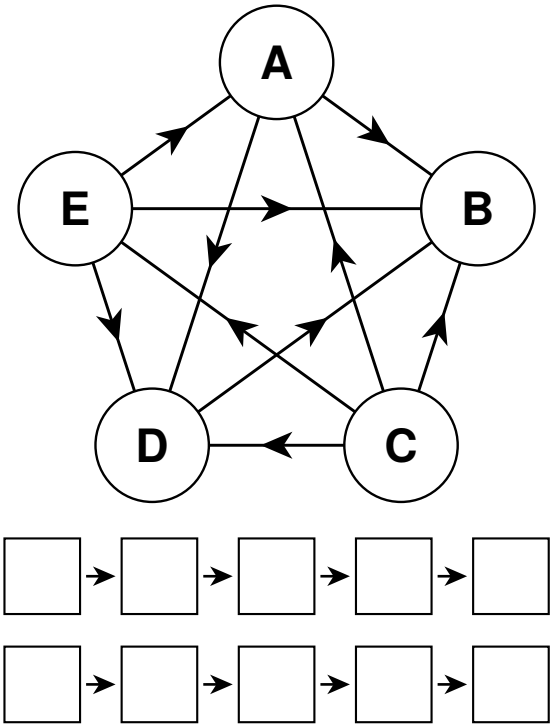


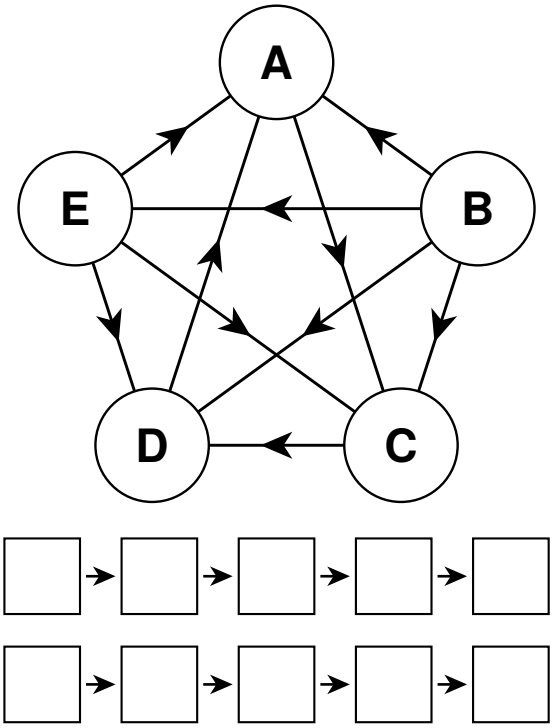


Problem 2: Find a line-up for each tournament. Which of these tournaments have only one line-up?

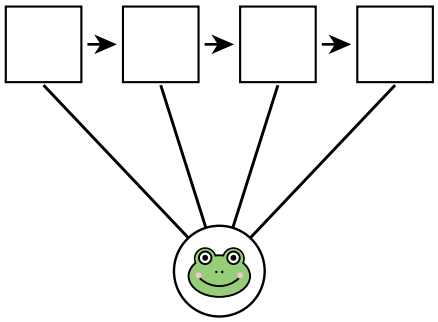


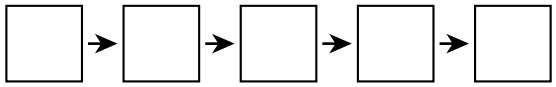


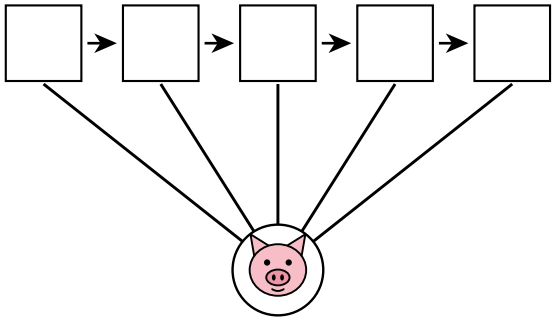


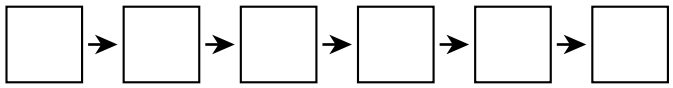


Problem 3: The frog joins your last tournament from Problem 1 and plays the other four animals. Copy your line-up above the frog and draw the frog's games. Then fit the frog into the line-up without changing the order of the other four. Do the same when the pig joins and plays the other five.

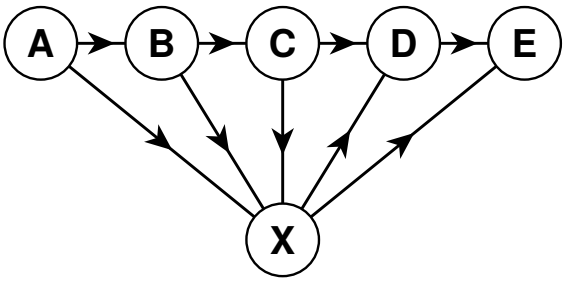


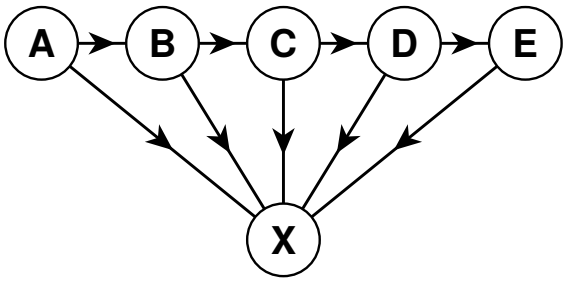


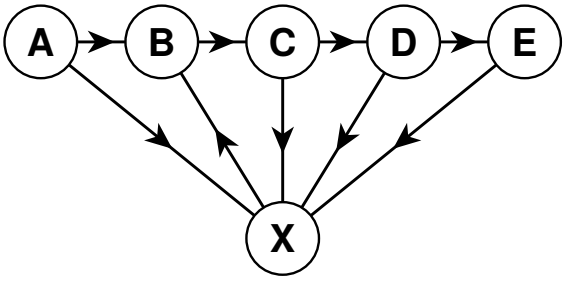


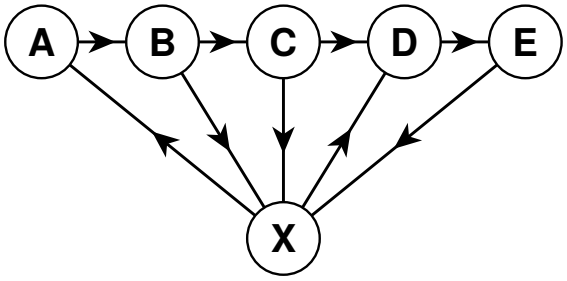


Problem 4: Player X plays each player in the line-up A, B, C, D, E. In each picture, mark every place where X can join the line-up without changing the order of the others.

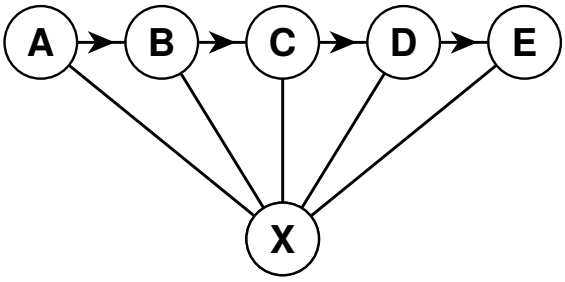


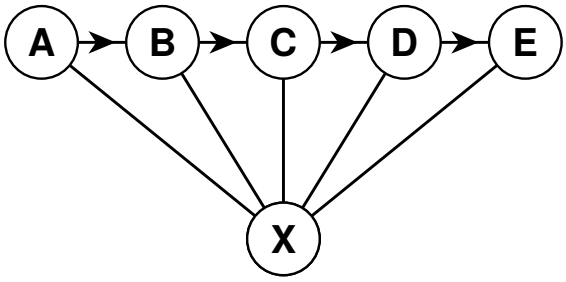




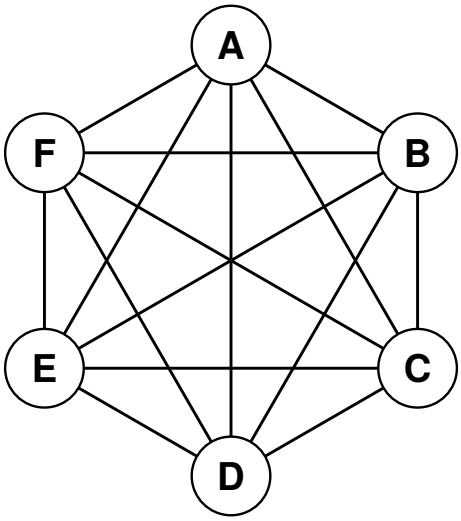


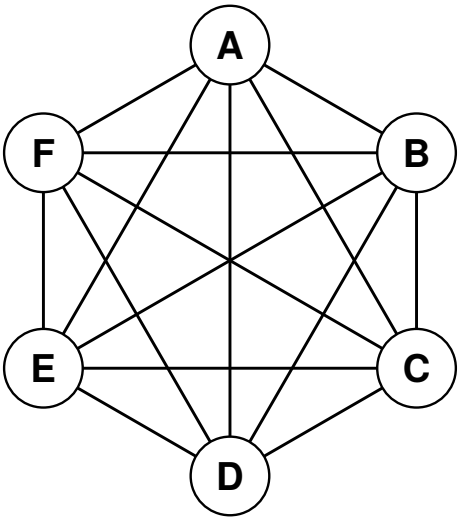
Problem 5: Draw arrowheads on the lines to X so that X cannot join the line-up A, B, C, D, E anywhere, or explain why that is impossible.



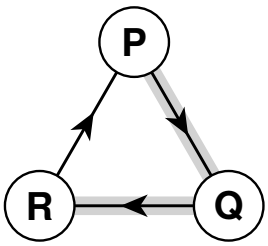


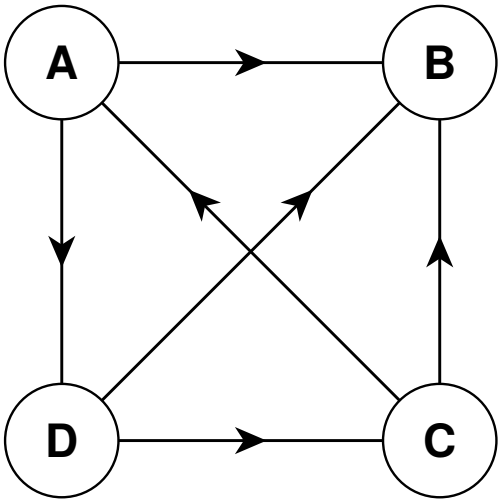
Problem 6: Make a tournament of six players with no line-up, or explain why there is no such tournament.

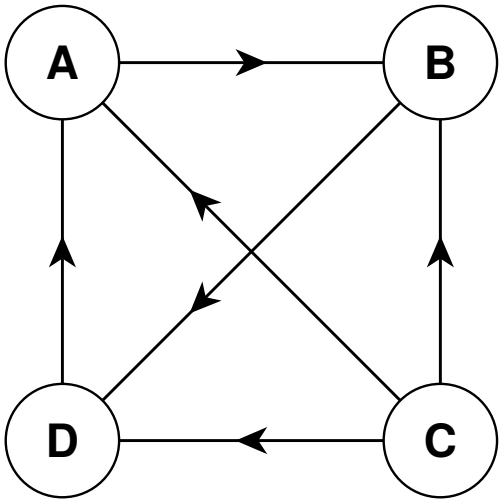


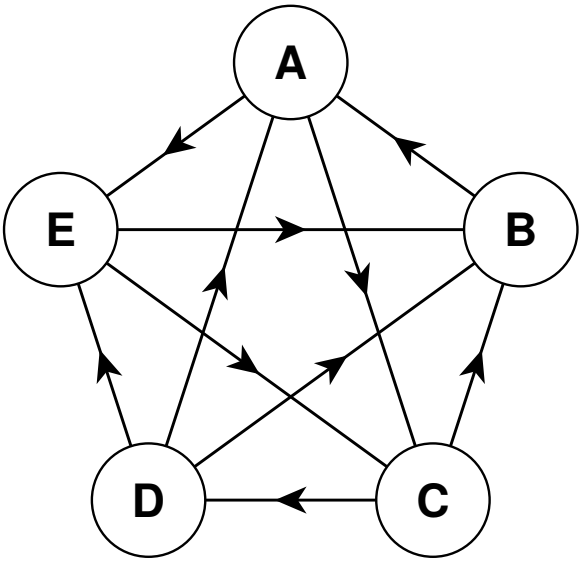


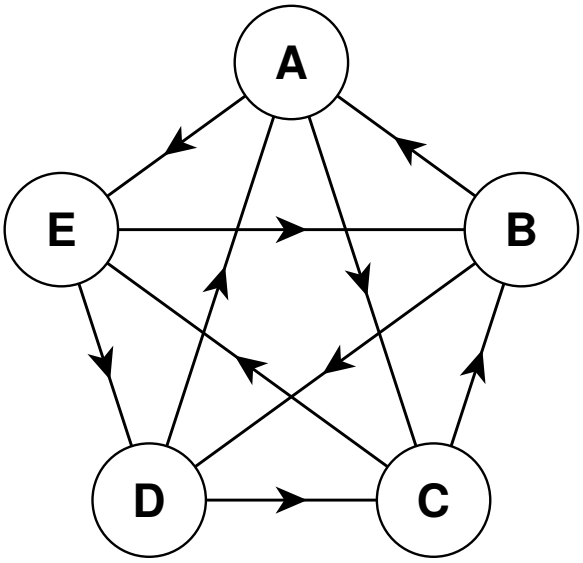
Problem 7: A player is a king if you can get from that player to every other player by following one arrow or two arrows. Here P gets to Q by one arrow and to R by two arrows. Find all the kings in each tournament.



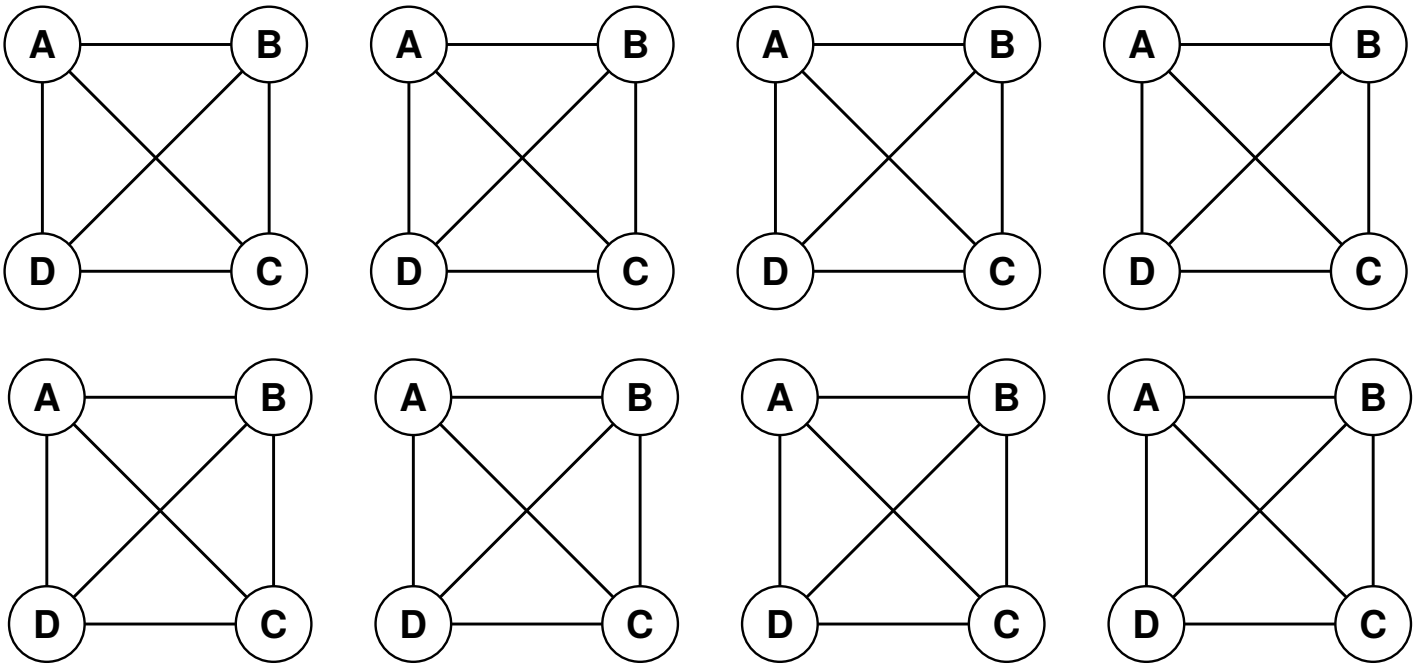








Problem 8: Try to make a tournament of four players with exactly 1 king, one with exactly 2 kings, one with exactly 3 kings, and one with exactly 4 kings. Which of these can be made?



Problem 9: Is a player who won the most games always a king? Explain.